

A little introduction to matrix calculus
Familiar form from 1st semester of calculus:

$$y = f(x), \quad \frac{dy}{dx} = \frac{d}{dx} f(x).$$

↑
scalar
function
(returns a
scalar)

Ex: $y = mx + b$

$$\frac{dy}{dx} = \frac{d}{dx} (mx + b) = m$$

What if x is a vector?

scalar vector
↑ ↑
 $y = f(\vec{x}) = f(x_1, \dots, x_n)$

$$\frac{dy}{d\vec{x}} = \frac{df(\vec{x})}{d\vec{x}}$$

What does it mean to take the derivative with respect to a vector?

The traditional way to handle this is to compute the derivative ~~is each~~ by finding the direction of steepest ascent. The magnitude of the vector tells us how steep

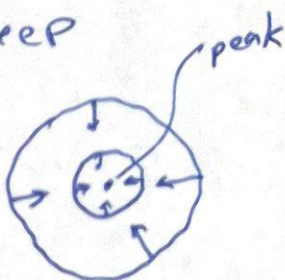


Fig 1: contour plot representing a hill. Each vector points uphill. The magnitude represents the steepness

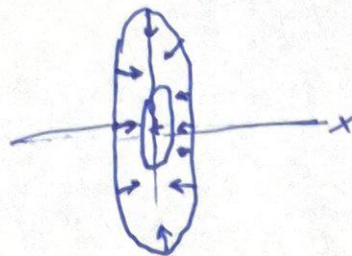


Fig 2: steepest ascent doesn't always point toward the peak. In the example above steepest ascent only points at the peak when approaching along the x or y axes. steepest ascent is orthogonal to the contour curve.

The vector pointing in the direction of steepest ascent is known as the gradient. We use ∇ (ak.a., "nabla") to denote the gradient

$\nabla_{\bar{x}} f(\bar{x}) \leftarrow$ The gradient of the scalar function f with respect to $\bar{x}$.

The gradient returns a vector

$$\nabla_{\bar{x}} f(\bar{x}) = \left(\frac{\partial}{\partial x_1} f(\bar{x}), \frac{\partial}{\partial x_2} f(\bar{x}), \dots, \frac{\partial}{\partial x_n} f(\bar{x}) \right)$$

In matrix form this is written

$$\nabla_{\bar{x}} f(\bar{x}) = \underbrace{\begin{bmatrix} \frac{\partial}{\partial x_1} f(\bar{x}) \\ \frac{\partial}{\partial x_2} f(\bar{x}) \\ \vdots \\ \frac{\partial}{\partial x_n} f(\bar{x}) \end{bmatrix}}_{\text{column vector}}$$

For example, $y = f(\bar{x}) = \bar{a}^T \bar{x}$

$$[a_1 \ a_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

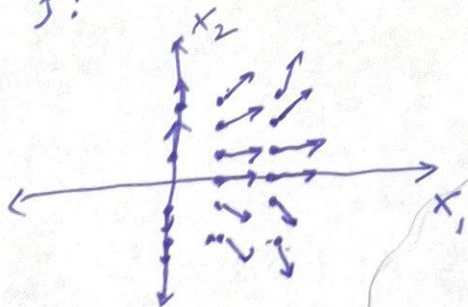
$$\nabla_{\bar{x}} (\bar{a}^T \bar{x}) = \nabla_{\bar{x}} (a_1 x_1 + a_2 x_2) = \begin{bmatrix} \frac{\partial}{\partial x_1} (a_1 x_1 + a_2 x_2) \\ \frac{\partial}{\partial x_2} (a_1 x_1 + a_2 x_2) \end{bmatrix}$$

This generalises to vectors w/ n dimensions

$$\boxed{\nabla_{\bar{x}} \bar{a}^T \bar{x} = \bar{a}}$$

What if $\vec{f}(\vec{x})$ is a vector function, meaning $\vec{f}(\vec{x})$ returns a vector? This is common in physics, ^{EE} or mechanical engineering.

f :



$f: X \rightarrow \mathbb{R}^n$
 $\nwarrow$ vector
 $\nwarrow$ any domain. Can be scalar, vector, or matrix.

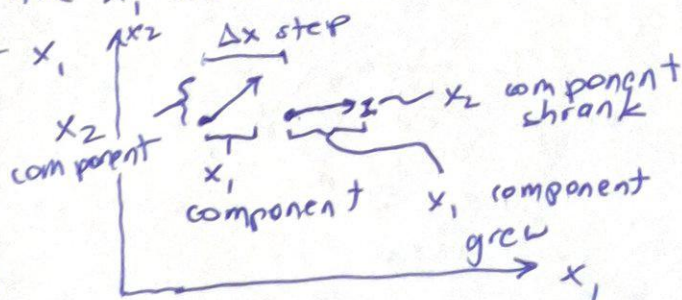
Figure 2: we define a vector for every point in space.

We think of the derivative as the change in the vectors' direction and magnitude as we step in any direction.

$$\nabla_{\vec{x}} \vec{f}(\vec{x}) = \left[\begin{array}{ccc} \frac{\partial}{\partial x_1} f_1 & \frac{\partial}{\partial x_2} f_1 & \dots & \frac{\partial}{\partial x_n} f_1 \\ \frac{\partial}{\partial x_1} f_2 & \dots & \dots & \frac{\partial}{\partial x_n} f_2 \\ \vdots & & & \\ \frac{\partial}{\partial x_1} f_n & \dots & \dots & \frac{\partial}{\partial x_n} f_n \end{array} \right] \left. \vphantom{\begin{array}{ccc} \frac{\partial}{\partial x_1} f_1 & \frac{\partial}{\partial x_2} f_1 & \dots & \frac{\partial}{\partial x_n} f_1 \end{array}} \right\} \begin{array}{l} \text{called} \\ \text{the} \\ \text{Jacobian} \\ \uparrow \\ \text{sometimes} \\ \text{denote} \\ J_{\vec{f}}(\vec{x}) \end{array}$$

$\frac{\partial}{\partial x_1} f_1$ is conceptually the change in the magnitude of component of $\vec{f}$ in the x_1 direction when we step in the direction of x_1 .

$\frac{\partial}{\partial x_1} f_2$ is the change of $\vec{f}$ in the x_2 direction when we step in the x_1 direction.



Example vector function

$$\bar{y} = f(\bar{x}) \quad f: X \rightarrow \mathbb{R}^n$$

consider the case when X is a column vector $n \times 1$.
 then $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$.
 Let $f(x)$ be a linear vector function.

$$\bar{y} = A \bar{x}$$

$n \times n$ matrix dimensions $n \times 1$ column vector

$$(n \times n) \cdot (n \times 1)$$

matrix multiply

The inner dimensions must be equal.

The output dimensionality is always the first dimension and the last dimension

The output $\bar{y}$ is an $n \times 1$ column vector.

Let $n=2$.

$$\bar{y} = A \bar{x} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\nabla_{\bar{x}} (A \bar{x}) = \nabla_x \left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

$$= \nabla_x \left(\begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix} \right)$$

Jacobian Example (cont.)

$$\nabla_{\bar{x}} \bar{f}(x) = \begin{bmatrix} \frac{\partial}{\partial x_1} (a_{11}x_1 + a_{12}x_2) & \frac{\partial}{\partial x_2} (a_{11}x_1 + a_{12}x_2) \\ \frac{\partial}{\partial x_1} (a_{21}x_1 + a_{22}x_2) & \frac{\partial}{\partial x_2} (a_{21}x_1 + a_{22}x_2) \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = A$$

This also generalizes to $n \times n$ matrixes so

$$\bar{f}(x) = A\bar{x}$$

$$\nabla_{\bar{x}} \bar{f}(x) = \boxed{\nabla_{\bar{x}} (A\bar{x}) = A} \leftarrow \nabla_{\bar{x}} A\bar{x} = A$$

What is the analog to $y = ax^2$?

The version for $n \times 1$ column vectors is without a constant is

$$y = x_1^2 + x_2^2 + \dots + x_n^2$$

which is the same as

$$y = \bar{x} \cdot \bar{x} \leftarrow \text{dot product of } \bar{x} \text{ with itself}$$

$$y = \bar{x}^T \bar{x}$$

The form with constant

$$y = \bar{x}^T A \bar{x}$$

$$\nabla_{\bar{x}} y = \nabla_{\bar{x}} \bar{x}^T A \bar{x} = (A + A^T) \bar{x}$$

When A is symmetric $\nabla_{\bar{x}} y = 2A\bar{x}$

I'll leave showing this as a homework problem.